

Hao Shi

Research Scientist

Shanda AI Research Tokyo

Speech intelligence for complex acoustic scenes

hshi@ieee.org

sites.google.com/view/hshi-speech

Research profile

Research scientist working on **speech intelligence**, with a focus on **robust speech processing**, **multi-talker speech understanding**, and **speech language models**. My work spans speech perception, recognition, separation, and generation in challenging real-world acoustic environments.

RESEARCH DIRECTION

My research follows a common goal: maintaining reliable correspondence between acoustic observations and model outputs as speech scenes and model priors become increasingly complex. This leads from **reliable perception**, through **evidence attribution**, to **evidence-grounded modeling**.

01	RELIABLE PERCEPTION Robust Speech Processing	Recover reliable acoustic evidence from degraded observations. Speech Enhancement · Noise-Robust ASR · Dereverberation
02	EVIDENCE ATTRIBUTION Multi-Talker & Speaker Grounding	Attribute speech content and acoustic sources in complex multi-speaker scenes. Multi-Talker ASR · Speaker Grounding · Separation / TSE
03	GROUNDING MODELING Grounded Speech LLMs & Spoken Interaction	Leverage learned priors while keeping reasoning and generation faithful to speech evidence. Speech LLMs · Grounded Generation · Speech-to-Speech

RESEARCH EXPERIENCE

2026–Present	Research Scientist Shanda Group Corp <i>Research focus: speech recognition, generation, and grounded speech models</i>	Tokyo, Japan
2025–2026	Research Scientist SB Intuitions, SoftBank <i>Research focus: speech-to-speech modeling and spoken interaction</i>	Tokyo, Japan
2024–2025	Researcher Kyoto University <i>Research focus: multi-talker ASR and speaker-aware speech processing</i>	Kyoto, Japan
2024	Research Fellow Kyoto University <i>Research focus: noise-robust automatic speech recognition</i>	Kyoto, Japan
2023	Research Intern NTT, CS Lab @ Keihanna <i>Research focus: deterministic/diffusion speech enhancement and system fusion</i>	Kyoto, Japan
2023	Research Intern Sony R&D, Osaki <i>Research focus: diffusion-based speech enhancement</i>	Tokyo, Japan

EDUCATION

2021–2024	Ph.D. in Informatics Kyoto University Graduate School of Informatics	Kyoto, Japan
2018–2021	M.Sc. in Computer Science and Technology Tianjin University College of Intelligence and Computing	Tianjin, China
2014–2018	B.Sc. in Computer Science and Technology Southwest Jiaotong University School of Information Science and Technology	Sichuan, China

ACADEMIC SERVICE

Organizing	Special Session Organizer: “Speaker Grounding in Speech LLMs: Recognition, Attribution, and Understanding in Multi-Talker Scenes,” ICASSP 2027. Challenge Organizer: “Automatic Song Aesthetics Evaluation Challenge,” ICASSP 2026.
Chairing	Session Chair, ICASSP 2026.
Reviewing	IEEE/ACM TASLP, Speech Communication, IEEE SPL, ICASSP, INTERSPEECH, SLT, ASRU, WASPAA, and APSIPA ASC.

PUBLICATIONS

Journal Papers

33. Yuan Gao, **Hao Shi**, Yahui Fu, Chenhui Chu, and Tatsuya Kawahara, “Bridging Speech Emotion Recognition and Personality: Dataset and Temporal Interaction Condition Network,” *IEEE Transactions on Affective Computing*, vol. 17, no. 1, pp. 829–840, 2026.
32. **Hao Shi**, Xugang Lu, Kazuki Shimada, and Tatsuya Kawahara, “Combining Deterministic and Diffusion Model to Meet the Partially Stochastic Function Property of Speech Enhancement,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 33, pp. 4253–4266, 2025.
31. Yuan Gao, **Hao Shi**, Chenhui Chu, and Tatsuya Kawahara, “Multi-Attribute Learning for Multi-Level Emotion Recognition from Speech,” *APSIPA Transactions on Signal and Information Processing*, vol. 14, no. e20, pp. 1–29, 2025.
30. **Hao Shi**, Masato Mimura, and Tatsuya Kawahara, “Waveform-domain Speech Enhancement Using Spectrogram Encoding for Robust Speech Recognition,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 32, pp. 3049–3060, 2024.

Conference Papers

29. **Hao Shi**, Yusuke Fujita, Roman Koshkin, Mengjie Zhao, Yuan Gao, Lianbo Liu, and Yui Sudo, “Distilling LLM Semantic Priors into Encoder-Only Multi-Talker ASR with Talker-Count Routing,” in *Proc. INTERSPEECH*, 2026.
28. Haoyang Li, Changsong Liu, Wei Rao, **Hao Shi**, Sakriani Sakti, and Eng Siong Chng, “Training-Free Intelligibility-Guided Observation Addition for Noisy ASR,” in *Proc. INTERSPEECH*, 2026.
27. Mengjie Zhao, Lianbo Liu, Yusuke Fujita, **Hao Shi**, Yuan Gao, Roman Koshkin, and Yui Sudo, “Speech-Worthy Alignment for Japanese SpeechLLMs via Direct Preference Optimization,” in *Proc. INTERSPEECH*, 2026.
26. Atsushi Kojima, Yusuke Fujita, **Hao Shi**, Tomoya Mizumoto, Mengjie Zhao, and Yui Sudo, “Conversation Context-aware Direct Preference Optimization for Style-Controlled Speech Synthesis,” in *Proc. APSIPA ASC*, 2025, pp. 573–578.
25. **Hao Shi**, Yusuke Fujita, Tomoya Mizumoto, Lianbo Liu, Atsushi Kojima, and Yui Sudo, “Serialized Output Prompting for Large Language Model-based Multi-Talker Speech Recognition,” in *Proc. IEEE ASRU*, 2025.
24. Tomoya Mizumoto, Yusuke Fujita, **Hao Shi**, Lianbo Liu, Atsushi Kojima, and Yui Sudo, “Evaluating Japanese Dialect Robustness across Speech and Text-based Large Language Models,” in *Proc. IEEE ASRU*, 2025.
23. Jiahui Zhao, **Hao Shi**, Tianrui Wang, Hexin Liu, Zhaoheng Ni, Lingxuan Ye, and Longbiao Wang, “Adapting Pretrained Speech Recognition Models for Code-Switching through Encoding Refining and Language-Aware Attention-based Decoding,” in *Proc. IEEE ICASSP*, 2025.
22. Zhongjian Cui, Chenrui Cui, Tianrui Wang, Mengnan He, **Hao Shi**, Meng Ge, Caixia Gong, Longbiao Wang, and Jianwu Dang, “Reducing the Gap between Pretrained Speech Enhancement and Recognition Models Using a Real Speech-Trained Bridging Module,” in *Proc. IEEE ICASSP*, 2025.
21. **Hao Shi**, Yuan Gao, Zhaoheng Ni, and Tatsuya Kawahara, “Serialized Speech Information Guidance with Overlapped Encoding Separation for Multi-Speaker Automatic Speech Recognition,” in *Proc. IEEE SLT*, 2024, pp. 193–199.
20. **Hao Shi**, and Tatsuya Kawahara, “Dual-path Adaptation of Pretrained Feature Extraction Module for Robust Automatic Speech Recognition,” in *Proc. INTERSPEECH*, 2024, pp. 2850–2854.
19. Yuan Gao, **Hao Shi**, Chenhui Chu, and Tatsuya Kawahara, “Speech Emotion Recognition with Multi-level Acoustic and Semantic Information Extraction and Interaction,” in *Proc. INTERSPEECH*, 2024, pp. 1060–1064.
18. Yuchun Shu, Bo Hu, Yifeng He, **Hao Shi**, Longbiao Wang, and Jianwu Dang, “Error Correction by Paying Attention to Both Acoustic and Confidence References for Automatic Speech Recognition,” in *Proc. INTERSPEECH*, 2024, pp. 3500–3504.
17. **Hao Shi**, Naoyuki Kamo, Marc Delcroix, Tomohiro Nakatani, and Shoko Araki, “Ensemble Inference for Diffusion Model-based Speech Enhancement,” in *Proc. IEEE ICASSPW*, 2024, pp. 735–739.
16. **Hao Shi**, Kazuki Shimada, Masato Hirano, Takashi Shibuya, Yuichiro Koyama, Zhi Zhong, Shusuke Takahashi, Tatsuya Kawahara, and Yuki Mitsufuji, “Diffusion-Based Speech Enhancement with Joint Generative and Predictive Decoders,” in *Proc. IEEE ICASSP*, 2024, pp. 12951–12955.
15. Yuan Gao, **Hao Shi**, Chenhui Chu, and Tatsuya Kawahara, “Enhancing Two-stage Finetuning for Speech Emotion Recognition Using Adapters,” in *Proc. IEEE ICASSP*, 2024, pp. 11316–11320.
14. Zhi Zhong, **Hao Shi**, Masato Hirano, Kazuki Shimada, Kazuya Tateishi, Takashi Shibuya, Shusuke Takahashi, and Yuki Mitsufuji, “Extending Audio Masked Autoencoders Toward Audio Restoration,” in *Proc. WASPAA*, 2023, pp. 1–5.
13. **Hao Shi**, Masato Mimura, Longbiao Wang, Jianwu Dang, and Tatsuya Kawahara, “Time-domain Speech Enhancement Assisted by Multi-resolution Frequency Encoder And Decoder,” in *Proc. IEEE ICASSP*, 2023, pp. 1–5.
12. Yanbing Yang, **Hao Shi**, Yuqin Lin, Meng Ge, Longbiao Wang, Qingzhi Hou, and Jianwu Dang, “Adaptive Attention

- Network with Domain Adversarial Training for Multi-Accent Speech Recognition,” in *Proc. ISCSLP*, 2022, pp. 6–10.
11. **Hao Shi**, Yuchun Shu, Longbiao Wang, Jianwu Dang, and Tatsuya Kawahara, “Fusing Multiple Bandwidth Spectrograms for Improving Speech Enhancement,” in *Proc. APSIPA ASC*, 2022, pp. 1935–1940.
 10. **Hao Shi**, Longbiao Wang, Sheng Li, Jianwu Dang, and Tatsuya Kawahara, “Subband-Based Spectrogram Fusion for Speech Enhancement by Combining Mapping and Masking Approaches,” in *Proc. APSIPA ASC*, 2022, pp. 286–292.
 9. **Hao Shi**, Longbiao Wang, Sheng Li, Jianwu Dang, and Tatsuya Kawahara, “Monaural speech enhancement based on spectrogram decomposition for convolutional neural network-sensitive feature extraction,” in *Proc. INTERSPEECH*, 2022, pp. 221–225.
 8. Tongtong Song, Qiang Xu, Meng Ge, Longbiao Wang, **Hao Shi**, Yongjie Lv, Yuqin Lin, and Jianwu Dang, “Language-specific Characteristic Assistance for Code-switching Speech Recognition,” in *Proc. INTERSPEECH*, 2022, pp. 3924–3928.
 7. Qiang Xu, Tongtong Song, Longbiao Wang, **Hao Shi**, Yuqin Lin, Yongjie Lv, Meng Ge, Qiang Yu, and Jianwu Dang, “Self-Distillation Based on High-level Information Supervision for Compressing End-to-End ASR Model,” in *Proc. INTERSPEECH*, 2022, pp. 1716–1720.
 6. **Hao Shi**, Longbiao Wang, Sheng Li, Cunhang Fan, Jianwu Dang, and Tatsuya Kawahara, “Spectrograms Fusion-based End-to-end Robust Automatic Speech Recognition,” in *Proc. APSIPA ASC*, 2021, pp. 438–442.
 5. Luya Qiang, **Hao Shi**, Meng Ge, Haoran Yin, Nan Li, Longbiao Wang, Sheng Li, and Jianwu Dang, “Speech Dereverberation Based on Scale-aware Mean Square Error Loss,” in *Proc. ICONIP*, 2021, pp. 55–63.
 4. Haoran Yin, **Hao Shi**, Longbiao Wang, Luya Qiang, Sheng Li, Meng Ge, Gaoyan Zhang, and Jianwu Dang, “Simultaneous Progressive Filtering-based Monaural Speech Enhancement,” in *Proc. ICONIP*, 2021, pp. 213–221.
 3. **Hao Shi**, Longbiao Wang, Meng Ge, Sheng Li, and Jianwu Dang, “Spectrograms Fusion with Minimum Difference Masks Estimation for Monaural Speech Dereverberation,” in *Proc. IEEE ICASSP*, 2020, pp. 7544–7548.
 2. **Hao Shi**, Longbiao Wang, Sheng Li, Chenchen Ding, Meng Ge, Nan Li, Jianwu Dang, and Hiroshi Seki, “Singing Voice Extraction with Attention based Spectrograms Fusion,” in *Proc. INTERSPEECH*, 2020, pp. 2412–2416.
 1. Meng Ge, Longbiao Wang, Nan Li, **Hao Shi**, Jianwu Dang, and Xiangang Li, “Environment-dependent attention-driven recurrent convolutional neural network for robust speech enhancement,” in *Proc. INTERSPEECH*, 2019, pp. 3153–3157.

Reports

- Lianbo Liu, Shiao Zhu, Kai Washizaki, Reo Yoneyama, Haesung Jeon, Mengjie Zhao, Yusuke Fujita, **Hao Shi**, Nao Yoshida, Yuan Gao, Roman Koshkin, Yukiya Hono, and Yui Sudo, “Sarashina2.2-TTS: Tackling Kanji Polyphony in Japanese Speech Generation via Data Scaling and Targeted Data Synthesis,” arXiv preprint arXiv:2606.25369, 2026.
- **Hao Shi**, and Tatsuya Kawahara, “Investigation of Adapter for Automatic Speech Recognition in Noisy Environment,” SIG Technical Reports, 2023, pp. 1–6.